LIDAR1
3D LiDAR
(Mid-360-class)

CAM-F
front camera

CAM-R
rear camera

GNSS1 F9P
rear-left mast
(RTK rover)

GNSS2 F9P
rear-right mast
(heading, 1 m)

IMU1
tilt / roll
(I²C)

RF1 LoRa 868 MHz
pendant + remote
stop + heartbeat

ETH (PoE / M12-X)

USB

RTR1
4G router
(SIM)

ETH

SW1
Ethernet switch
5-port 12/24 V

ETH

PC1 main computer
Jetson Orin Nano-class
ROS 2: navigation,
mission, logging

USB + CAN

MCU
motion + I/O controller
(STM32-class on
24 V I/O board)
watchdog to PC1

RS485

WG1 load-cell amp
4 ch, RS485
← LC1…LC4

HMI1 display
weight / state
(RS485)

4G: telemetry,
RTK corrections

CAN / RS485 / RC lead to the base (whatever it offers)

24 V I/O

868 MHz

BASE STATION (farm yard)
mini PC: fleet manager, logs
RTK reference F9P → NTRIP
certified scale (RS232)
4G / Wi-Fi router

BASE (bought, not modified)
motor controllers L/R, battery + BMS,
motors + brakes, RC receiver, E-stop input
← CAN / RS485, or RC-PWM imitation via K9

MCU I/O (24 V)
OUT: K4 watchdog, K5 drive en,
K7/K8 lift, K9 RC changeover,
beacon, buzzer
IN: lift limits S5/S6, base
charging, KS status, E-stop aux

PENDANTS (harvesters)
CALL / SEND / STOP
LoRa
+ wireless E-stop TX
(→ WES, E-02)

BINBOT — autonomous olive bin carrier

CONTROL AND COMMUNICATION — BLOCK DIAGRAM

DWG E-03 REV C BP-04 electronics

PRELIMINARY — check against vendor datasheets 2026-10-28